

Xuanru Zhou

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EDUCATION

Zhejiang University

B.A. College of Electrical and Engineering, Automation; **GPA: 3.95/4.00**

Minor in Chu Kochen Honors College, Advanced Honor Class of Engineering Education (ACEE) Apr 2022 – Jun 2023

Hangzhou, China

Sep 2021 – Jun 2025

EXPERIENCE

Berkeley Speech Group, UC Berkeley

Remote Research Intern

Mar 2025 – Present

- Exploring speech, audio-visual world model, multimodal reasoning, grounded language & speech intelligence in collaboration with PhD. Jiachen Lian.

Tencent AI Lab

Research Intern

Shenzhen, China

Apr 2025 – Present

- Developing contrastive learning algorithms for audio data pretraining, scaling audio-text training across various tasks to explore and bridge the modality gap for general audio understanding.
- Hands-on experience in developing and training large-scale spoken language models by fine-tuning LLMs with speech encoder front-ends.

Berkeley Speech Group, UC Berkeley

Visiting Student Researcher

Berkeley, CA

Sep 2024 – Feb 2025

- Worked on verbatim speech transcription model for speech pronunciation assessment, proposed phoneme similarity modeling method that markedly improved the overall performance.

Berkeley Speech Group, UC Berkeley

Remote Research Intern

Aug 2023 – Aug 2024

- Developed a multi-lingual, end-to-end model for region-wise (time-based) detection of speech dysfluency.
- Proposed a novel token-based dysfluency detection method, in conjunction with the time-based model, established benchmark for the field.

Networked Sensing and Control Group, Zhejiang University (NeSC)

Student Research Assistant

Hangzhou, China

Apr 2023 – Aug 2024

- Developed an audio watermarking paradigm to protect voice copyright.
- Worked on dataset copyright auditing in machine learning systems.

AWARDS

Zhejiang Provincial Government Scholarship

Oct 2022

Zhejiang University Scholarship

Oct 2022, Oct 2023

ISCA Student Travel Award

Jun 2024

SLT 2024 Student Travel Grant

Sep 2024

NeurIPS 2024 Scholar Award

Oct 2024

Outstanding Undergraduate Graduate of Zhejiang University

Jun 2025

Outstanding Undergraduate Thesis Award

Jun 2025

SELECTED PUBLICATIONS

Speech World Model: Causal State–Action Planning with Explicit Reasoning for Speech.

ICLR 2026 in submission.

Xuanru Zhou, Jiachen Lian, Henry Hong, Xinyi Yang, Gopala Anumanchipalli

Revisiting Audio-language Pretraining for Learning General-purpose Audio Representation.

ICLR 2026 in submission.

Wei-Cheng Tseng*, Xuanru Zhou*, Mingyue Huo, Yiwen Shao, Hao Zhang, Dong Yu

SoK: Dataset Copyright Auditing in Machine Learning Systems. IEEE S&P 2025

Linkang Du*, Xuanru Zhou*, Min Chen, Zhou Su, Peng Cheng, Jiming Chen, Zhikun Zhang

Towards Accurate Phonetic Error Detection Through Phoneme Similarity Modeling. Interspeech 2025

Xuanru Zhou, Jiachen Lian, Cheol Jun Cho, Tejas Prabhune, Shuhe Li, William Li, Rodrigo Ortiz, Zoe Ezzes, Jet Vonk, Brittany Morin, Rian Bogley, Lisa Wauters, Zachary Miller, Maria Luisa Gorno-Tempini, Gopala Anumanchipalli

SSDM: Scalable Speech Dysfluency Modeling. NeurIPS 2024

Jiachen Lian, Xuanru Zhou, Zoe Ezzes, Jet Vonk, Brittany Morin, David Paul Galang Baquirin, Zachary Miller, Maria Luisa Gorno-Tempini, Gopala Anumanchipalli

Automatic Detection of Articulatory-Based Disfluencies in Primary Progressive Aphasia. IEEE JSTSP 2025

Jiachen Lian, Xuanru Zhou, Zoe Ezzes, Jet Vonk, Brittany Morin, David Paul Galang Baquirin, Zachary Miller, Maria Luisa Gorno-Tempini, Gopala Anumanchipalli

YOLO-Stutter: End-to-end Region-Wise Speech Dysfluency Detection. Interspeech 2024

Xuanru Zhou, Anshul Kashyap, Steve Li, Ayati Sharma, Brittany Morin, David Baquirin, Jet Vonk, Zoe Ezzes, Zachary Miller, Maria Luisa Gorno-Tempini, Jiachen Lian, Gopala Krishna Anumanchipalli

Stutter-Solver: End-to-end Multi-lingual Dysfluency Detection. IEEE SLT 2024

Xuanru Zhou, Cheol Jun Cho, Ayati Sharma, Brittany Morin, David Baquirin, Jet Vonk, Zoe Ezzes, Zachary Miller, Boon Lead Tee, Maria Luisa Gorno-Tempini, Jiachen Lian, Gopala Krishna Anumanchipalli

Time and Tokens: Benchmarking End-to-End Speech Dysfluency Detection. Preprint

Xuanru Zhou, Jiachen Lian, Cheol Jun Cho, Jingwen Liu, Zongli Ye, Jinming Zhang, Brittany Morin, David Baquirin, Jet Vonk, Zoe Ezzes, Zachary Miller, Maria Luisa Gorno-Tempini, Gopala Krishna Anumanchipalli

PROJECTS

Generative AIGC Technology from Ancient Chinese Poetry to Painting

- Collaborating to finetune the Chinese-CLIP model realizes matching of ancient poetry and ancient paintings to achieve understanding, supervised by Prof. Yingcai Wu.

ARP Attacks

- Using VMware virtual machine and Kali Linux operating system as the attack machine, physical host Win11 as target machine to implement ARP disconnection, ARP spoofing and other attacks.

Reversi - Mini AlphaGo

- A Python project in which human can play Reversi against an AI with the Monte Carlo tree search and Upper Confidence Bound1 algorithm.

SKILLS

Programming Languages: C/C++, Python, HTML, CSS, JavaScript, TypeScript, MATLAB

Tools & Frameworks: Git, Docker, AWS, React, PyTorch, TensorFlow

Languages: English, Chinese